

4.1 Answer each of the following:

- Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

a. $SST = \sum (y_i - \bar{y})^2 = \sum (y_i^2 - 2\bar{y} \sum y_i + \bar{y}^2) = \sum y_i^2 - 2 \times N \bar{y}^2 + N \bar{y}^2$
 $R^2 = SSR / SST = \frac{375.47}{445.458} = 0.8429$ $= 7825.94 - 20 \times 19.21^2 = 445.458$

b. $SSR = 0.7911 \times 725.94 = 574.291$
 $SSE = 151.6489$
 $df \text{ of } \hat{\sigma}^2 = 18 \quad \hat{\sigma}^2 = MSE = \frac{151.6489}{18} = 8.4249$

c. $SST = 631.63$
 $\sum \hat{e}_i^2 = \sum (y_i - \hat{y}_i)^2 = SSE = 182.85$
 $SSR = 448.78$
 $R^2 = \frac{448.78}{631.63} = 0.7105$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

$N = 5$
 $SST = 10$
 $SSE = 3.6$
 $MSE = 0.72$
 $\bar{x} = 1$

$SST_x = 10$

(a). $y | x_0 = 4 = 1.2 + 0.8 \times 4 = 4.4$

(b) $Var(y | x_0 = 4)$
 $= \sigma^2 \left(1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right)$
 $= 0.72 \times \left(1 + \frac{1}{5} + \frac{3^2}{10} \right) = 15.12$
 $se(f) = 3.888, \text{ at } x_0 = 4$

$\varphi = \frac{y_0 - \hat{y}_0}{se(f)} \sim t(N-2) = t(3)$

(c) $y | x_0 = 4 \in [\hat{y}_0 \pm t_{2.975}(3) \times se(f)]$
 $= [4.4 \pm 3.182 \times \sqrt{15.12}]$

$$= [-7.973, 16.773]$$

$$(D) y|x_0=4 \in [\hat{y}_0 \pm t_{0.5}(3) \times \text{se}(f)] \\ = [4.4 \pm 5.84 \sqrt{15.12}] \\ = [-18.312, 27.1124]$$

$$(E) y|x_0=\bar{x}=2$$

$$\text{se}(f) = \sqrt{\sigma^2 \left(1 + \frac{1}{5} + 0\right)} = \sqrt{7 \times 1.2} = 2.939$$

we find $\text{se}(f)$ is lower compared to $\text{se}(f|x_0=4)$ so the width is also lower and the C.I. is tighter.

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

- Estimate the log-linear model $\ln(\text{PRICE}) = \beta_1 + \beta_2 \text{SQFT} + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- Estimate the log-log model $\ln(\text{PRICE}) = \alpha_1 + \alpha_2 \ln(\text{SQFT}) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.
- Compare the R^2 value from the linear model $\text{PRICE} = \delta_1 + \delta_2 \text{SQFT} + e$ to the “generalized” R^2 measure for the models in (b) and (c).
- Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?
- For each of the models in (a)–(c), plot the least squares residuals against *SQFT*. Do you observe any patterns?
- For each model in (a)–(c), predict the value of a house with 2700 square feet.
- For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.
- Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

$$(a) \text{ slope} = b_1 = 0.036$$

$$\text{elasticity} = \frac{d(\ln(\text{price}))}{d \text{sqft}} \times \text{sqft} = 0.983$$

log-linear model

when *sqft* ↑ a unit,
ln(price) will ↑ 0.036 unit.

```
> summary(modelA)

Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
    Min       1Q   Median       3Q      Max
-1.32528 -0.19199  0.00122  0.21678  0.86609

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.393866   0.043288  101.50   <2e-16 ***
sqft         0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,    Adjusted R-squared:  0.5408
F-statistic: 588.7 on 1 and 498 DF, p-value: < 2.2e-16
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$$(b) \text{ slope} = \alpha_2 = 1.02$$

$$\text{elasticity} = \alpha_2 = 1.02$$

log-log model

when *sqft* ↑ 1%
price ↑ 1.025%

```
> summary(modelB)

Call:
lm(formula = log(price) ~ log(sqft), data = collegetown)

Residuals:
    Min       1Q   Median       3Q      Max
-1.36864 -0.20849  0.00487  0.23204  1.21099

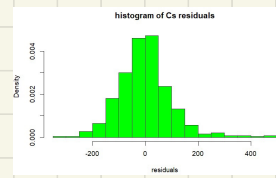
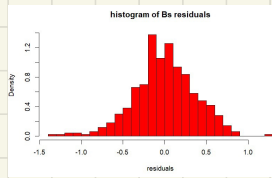
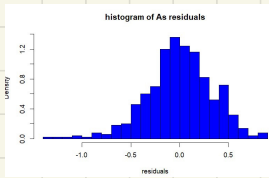
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.04965   0.15797   12.97   <2e-16 ***
log(sqft)    1.02483   0.04839   21.18   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3643 on 498 degrees of freedom
Multiple R-squared:  0.4738,    Adjusted R-squared:  0.4728
F-statistic: 448.5 on 1 and 498 DF, p-value: < 2.2e-16
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(c) $R^2_C = 0.6413$

$R^2_B = 0.9178$

(d)



a bit positively skewed seems to be okay (left skewed).
but ϵ has fat tail in these three models.

JB Test. $H_0: \epsilon \sim N(0, \sigma^2)$; $H_1: \epsilon$ doesn't follow normal distribution

Jarque Bera Test
data: resid(modelA)
X-squared = 26.679, df = 2, p-value = 1.61e-06

$p\text{-value} < \alpha = 0.025$

$\therefore H_0$ is rejected in model A

Jarque Bera Test
data: resid(modelB)
X-squared = 14.279, df = 2, p-value = 0.0007933

$p\text{-value} < \alpha = 0.025$

$\therefore H_0$ is rejected in model B

but in model B, we can say
it's skewness is not that large.

Jarque Bera Test
data: resid(modelC)
X-squared = 221.11, df = 2, p-value < 2.2e-16

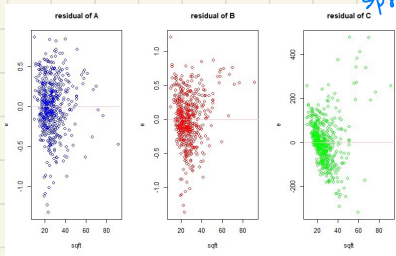
$p\text{-value} < \alpha = 0.025$

$\therefore H_0$ is rejected in model C

three models suggest that ϵ doesn't follow normal distribution

(e)

3个模型的残差都有漏斗的形状. 怀疑有异质变量和
spray.
heteroscedasticity.



(f) $sqft = 270$

模型A预测的房价: 214.2336
模型B预测房价: 227.5386
模型C预测房价: 246.4557

(g)

模型A预测的房价区间: 214.2336 109.7685 418.1167
模型B预测房价区间: 227.5386 111.1406 465.8406
模型C预测房价区间: 246.4557 44.27727 448.6341

(h)

We see that there is heteroscedasticity
problem in these three model
so we might need to find another model
to fit the assumption & use different
hypothesis test method.